

Michael (Misha, Mykhailo) Rashkovetskyi

CCAPP Postdoctoral Fellow (cosmology)
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Employment

- **The Ohio State University** Columbus, OH, USA
CCAPP Postdoctoral Fellow 2025 – 2028
– [Center for Cosmology and AstroParticle Physics](#)
- **Harvard College Observatory** Cambridge, MA, USA
Postdoctoral Fellow 2025

Education

- **Harvard University** Cambridge, MA, USA
Ph.D. in Astronomy and Astrophysics 2020 – 2025
M.A. in Astronomy and Astrophysics, in passing 2022
– Thesis: [Enhancing the analysis of the large-scale structure of the Universe for cutting-edge cosmological surveys with two-point correlation function and beyond](#) (May 2025)
– Advisor: Prof. Daniel Eisenstein
– [Center for Astrophysics | Harvard & Smithsonian](#)
- **Tel Aviv University** Tel Aviv-Yafo, Israel
B.Sc. in Physics, Summa Cum Laude (GPA: 98/100) 2019 – 2020
– Raymond & Beverly Sackler School of Physics & Astronomy
– Advisor: Dr. Omer Bromberg
- **Moscow Institute of Physics and Technology** Dolgoprudny, Russia
B.Sc. in Applied Mathematics and Physics, unfinished 2015 – 2018
– Department of General and Applied Physics
– Advisor: Prof. Vasily Beskin
- **Richelieu Lyceum** Odesa, Ukraine
High school, specialization in physics 2010 – 2015

First-author publications

4. M. **Rashkovetskyi**, D. J. Eisenstein, *et al.*, “Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect”, *The Open Journal of Astrophysics* **8** October (2025)a 46033, [arXiv:2508.20904](#)
3. M. **Rashkovetskyi**, D. Forero-Sánchez, A. de Mattia, D. J. Eisenstein, N. Padmanabhan, H. Seo, A. J. Ross, *et al.*, “Semi-analytical covariance matrices for two-point correlation function for DESI 2024 data”, *J. Cosmology Astropart. Phys.* **2025** January (2025)b 145, [arXiv:2404.03007](#)

2. M. **Rashkovetskyi**, D. J. Eisenstein, *et al.*, “Validation of semi-analytical, semi-empirical covariance matrices for two-point correlation function for early DESI data”, MNRAS **524** September (2023) 3894–3911, [arXiv:2306.06320](#)
1. M. **Rashkovetskyi**, J. B. Muñoz, D. J. Eisenstein, and C. Dvorkin, “Small-scale clumping at recombination and the Hubble tension”, Phys. Rev. D **104** November (2021) 103517, [arXiv:2108.02747](#)

First-tier author publications

10. E. A. Zaborowski, P. Taylor, K. Honscheid, A. Cuceu, A. de Mattia, A. Krolewski, M. **Rashkovetskyi**, A. J. Ross, C. To, *et al.*, “ H_0 without the sound horizon (or supernovae): A 2% measurement in DESI DR1”, J. Cosmology Astropart. Phys. **2026** April (2026) 004, [arXiv:2510.19149](#)
9. D. Valcin, M. **Rashkovetskyi**, H. Seo, F. Beutler, P. McDonald, A. de Mattia, A. J. Rosado-Marín, A. J. Ross, N. Padmanabhan, *et al.*, “Combined tracer analysis for DESI 2024 BAO”, J. Cosmology Astropart. Phys. **2026** April (2026) 058, [arXiv:2508.05467](#)
8. U. Andrade, E. Paillas, J. Mena-Fernández, Q. Li, A. J. Ross, S. Nadathur, M. **Rashkovetskyi**, A. Pérez-Fernández, H. Seo, N. Sanders, O. Alves, X. Chen, N. Deiosso, A. de Mattia, M. White, *et al.*, “Validation of the DESI DR2 measurements of baryon acoustic oscillations from galaxies and quasars”, Phys. Rev. D **112** October (2025) 083512, [arXiv:2503.14742](#)
7. D. Forero-Sánchez, M. **Rashkovetskyi**, O. Alves, *et al.*, “Analytical and EZmock covariance validation for the DESI 2024 results”, J. Cosmology Astropart. Phys. **2025** April (2025) 055, [arXiv:2411.12027](#)
6. J. Yu, A. J. Ross, A. Rocher, O. Alves, A. de Mattia, D. Forero-Sánchez, J.-P. Kneib, A. Krolewski, T. Lan, M. **Rashkovetskyi**, *et al.*, “ELG spectroscopic systematics analysis of the DESI Data Release 1”, J. Cosmology Astropart. Phys. **2025** January (2025) 126, [arXiv:2405.16657](#)
5. J. Mena-Fernández, C. Garcia-Quintero, S. Yuan, B. Hadzhiyska, O. Alves, M. **Rashkovetskyi**, H. Seo, N. Padmanabhan, S. Nadathur, C. Howlett, S. Alam, A. Rocher, A. J. Ross, E. Sanchez, M. Ishak, *et al.*, “HOD-dependent systematics for luminous red galaxies in the DESI 2024 BAO analysis”, J. Cosmology Astropart. Phys. **2025** January (2025) 133, [arXiv:2404.03008](#)
4. C. Garcia-Quintero, J. Mena-Fernández, A. Rocher, S. Yuan, B. Hadzhiyska, O. Alves, M. **Rashkovetskyi**, H. Seo, N. Padmanabhan, S. Nadathur, C. Howlett, M. Ishak, L. Medina-Varela, P. McDonald, A. J. Ross, Y. Xie, X. Chen, A. Bera, *et al.*, “HOD-dependent systematics in Emission Line Galaxies for the DESI 2024 BAO analysis”, J. Cosmology Astropart. Phys. **2025** January (2025) 132, [arXiv:2404.03009](#)
3. J. Moon, D. Valcin, M. **Rashkovetskyi**, C. Saulder, *et al.*, “First detection of the BAO signal from early DESI data”, MNRAS **525** November (2023) 5406–5422, [arXiv:2304.08427](#)
2. E. M. Novoselov, V. S. Beskin, A. K. Galishnikova, M. M. **Rashkovetskyi**, and A. V. Biryukov, “Orthogonal pulsars as a key test for pulsar evolution”, MNRAS **494** April (2020) 3899–3911, [arXiv:2004.03211](#)
1. V. S. Beskin, A. K. Galishnikova, E. M. Novoselov, A. A. Philippov, and M. M. **Rashkovetskyi**, “So how do radio pulsars slow-down?”, in “Journal of Physics Conference Series”, vol. 932, p. 012012. December 2017

Collaboration publications

19. C. Garcia-Quintero, H. E. Noriega, A. de Mattia, A. Aviles, K. Lodha, D. Chebat, J. Rohlf, S. Nadathur, W. Elbers, *et al.*, “Cosmological implications of DESI DR2 BAO measurements in light of the latest ACT DR6 CMB data”, *Phys. Rev. D* **112** October (2025) 083529, [arXiv:2504.18464](#)
18. DESI Collaboration *et al.*, “DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints”, *Phys. Rev. D* **112** October (2025)a 083515, [arXiv:2503.14738](#)
17. DESI Collaboration *et al.*, “DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest”, *Phys. Rev. D* **112** October (2025)b 083514, [arXiv:2503.14739](#)
16. K. Lodha, R. Calderon, W. L. Matthewson, A. Shafieloo, M. Ishak, J. Pan, C. Garcia-Quintero, D. Huterer, G. Valogiannis, L. A. Ureña-López, N. V. Kamble, D. Parkinson, A. G. Kim, G. B. Zhao, J. L. Cervantes-Cota, J. Rohlf, F. Lozano-Rodríguez, J. O. Román-Herrera, *et al.*, “Extended dark energy analysis using DESI DR2 BAO measurements”, *Phys. Rev. D* **112** October (2025) 083511, [arXiv:2503.14743](#)
15. W. Elbers, A. Aviles, H. E. Noriega, D. Chebat, A. Menegas, C. S. Frenk, C. Garcia-Quintero, D. Gonzalez, M. Ishak, O. Lahav, K. Naidoo, G. Niz, C. Yèche, *et al.*, “Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape”, *Phys. Rev. D* **112** October (2025) 083513, [arXiv:2503.14744](#)
14. DESI Collaboration *et al.*, “Data Release 1 of the Dark Energy Spectroscopic Instrument”, *AJ* **171** May (2026) 285, [arXiv:2503.14745](#)
13. X. Chen, Z. Ding, E. Paillas, S. Nadathur, H. Seo, S. Chen, N. Padmanabhan, M. White, A. de Mattia, P. McDonald, A. J. Ross, A. Variu, A. Carnero Rosell, B. Hadzhiyska, M. M. S. Hanif, D. Forero-Sánchez, *et al.*, “Extensive analysis of reconstruction algorithms for DESI 2024 baryon acoustic oscillations”, November 2024, [arXiv:2411.19738](#)
12. DESI Collaboration *et al.*, “DESI 2024 II: sample definitions, characteristics, and two-point clustering statistics”, *J. Cosmology Astropart. Phys.* **2025** July (2025)a 017, [arXiv:2411.12020](#)
11. DESI Collaboration *et al.*, “DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars”, *J. Cosmology Astropart. Phys.* **2025** September (2025)b 008, [arXiv:2411.12021](#)
10. DESI Collaboration *et al.*, “DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements”, *J. Cosmology Astropart. Phys.* **2025** July (2025)c 028, [arXiv:2411.12022](#)
9. A. Pérez-Fernández, L. Medina-Varela, R. Ruggeri, M. Vargas-Magaña, H. Seo, N. Padmanabhan, M. Ishak, *et al.*, “Fiducial-cosmology-dependent systematics for the DESI 2024 BAO analysis”, *J. Cosmology Astropart. Phys.* **2025** January (2025) 144, [arXiv:2406.06085](#)
8. U. Andrade, J. Mena-Fernández, H. Awan, A. J. Ross, S. Brieden, J. Pan, A. de Mattia, *et al.*, “Validating the galaxy and quasar catalog-level blinding scheme for the DESI 2024 analysis”, *J. Cosmology Astropart. Phys.* **2025** January (2025) 128, [arXiv:2404.07282](#)
7. DESI Collaboration *et al.*, “DESI 2024 III: baryon acoustic oscillations from galaxies and quasars”, *J. Cosmology Astropart. Phys.* **2025** April (2025)a 012, [arXiv:2404.03000](#)
6. DESI Collaboration *et al.*, “DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest”, *J. Cosmology Astropart. Phys.* **2025** January (2025)b 124, [arXiv:2404.03001](#)

5. DESI Collaboration *et al.*, “DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations”, J. Cosmology Astropart. Phys. **2025** February (2025)c 021, [arXiv:2404.03002](#)
4. E. Paillas, Z. Ding, X. Chen, H. Seo, N. Padmanabhan, A. de Mattia, A. J. Ross, S. Nadathur, C. Howlett, *et al.*, “Optimal reconstruction of baryon acoustic oscillations for DESI 2024”, J. Cosmology Astropart. Phys. **2025** January (2025) 142, [arXiv:2404.03005](#)
3. S. F. Chen, C. Howlett, M. White, P. McDonald, A. J. Ross, H. J. Seo, N. Padmanabhan, *et al.*, “Baryon acoustic oscillation theory and modelling systematics for the DESI 2024 results”, MNRAS **534** October (2024) 544–574, [arXiv:2402.14070](#)
2. DESI Collaboration *et al.*, “Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument”, AJ **167** February (2024)a 62, [arXiv:2306.06307](#)
1. DESI Collaboration *et al.*, “The Early Data Release of the Dark Energy Spectroscopic Instrument”, AJ **168** August (2024)b 58, [arXiv:2306.06308](#)

Selected open source contributions (<https://github.com/misharash>)

- [RASCALC](#) C++ & Python
Fast semi-analytic covariance matrices library/program 2022 – 2026
- [RASCALC scripts](#) Python
DESI covariance matrix pipeline for 2-point correlation function (scripts) 2024 – 2026
- [PYCORR](#) Python
Library for 2-point correlation function estimation 2024 – 2025

Seminar talks

- [Ohio Cosmic Connections](#) Cincinnati, OH, USA
Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect May 4, 2026
- [University of Michigan Cosmology Group Seminar](#) Ann Arbor, MI, USA
Semi-analytical covariance matrices for DESI BAO measurements, and more information from tSZ-split clustering April 10, 2026
- [Ohio University Astro Seminar](#) Athens, OH, USA
Enhancing the analysis of the large-scale structure of the Universe October 8, 2025
- [CCAPP Fellows Symposium 2025](#) Columbus, OH, USA
Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect September 17, 2025
- [KIPAC Tea \(Stanford / SLAC\)](#) Menlo Park, CA, USA
Fast semi-analytical covariance matrices for 2-point correlation functions May 10, 2024
- [ITC Luncheon](#) Cambridge, MA, USA
Semi-analytic covariance matrices for 2PCF of DESI galaxies (on YouTube) September 28, 2023

Contributed conference talks

- **APS Global Physics Summit** Denver, CO, USA
Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect March 17, 2026
- **28th International Conference on Particle Physics & Cosmology** Pittsburgh, PA, USA
Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect October 16, 2025
- **APS Global Physics Summit** Anaheim, CA, USA
Clustering of DESI LRG selected based on ACT DR6 + Planck tSZ map March 18, 2025
- **IX Essential Cosmology for the Next Generation** Playa del Carmen, Mexico
Clustering of DESI galaxies selected based on ACT thermal SZ map December 5, 2024
- **Cosmology from Home 2023** online
First Detection of the BAO Signal from Early DESI Data (on YouTube) July 4, 2023
Co-presenter with Christoph Saulder, Jeongin Moon and David Valcin
- **APS April Meeting** Minneapolis, MN, USA
First Detection of the BAO Signal from Early DESI Data April 18, 2023

Poster presentations

- **Arizona Winter School on Multi-Probe Cosmology** Tucson, AZ, USA
Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect January 12 – 15, 2025
Second best poster prize
- **Frontiers in Cosmology and Gravitational Physics** Portsmouth, UK
Fast semi-analytical covariance matrices for 2PCF of galaxies and quasars May 20 – 23, 2024
- **VIII Essential Cosmology for the Next Generation** Playa del Carmen, Mexico
RascalC: Empirical 2PCF Covariance Matrices without Mocks Nov 30 – Dec 3, 2022
- **CMB-S4 Summer Meeting** online
Small-scale Clumping at Recombination and the Hubble Tension August 9 – 13, 2021
- **CMB-S4 Spring Meeting** online
Hubble Tension with Small-Scale Clumping March 8 – 12, 2021
- **Physics of Neutron Stars - 2017** Saint-Petersburg, Russia
On the light-curve anomalies of radio pulsars July 10 – 14, 2017

Other meetings

- **DESI December** Tucson, AZ, USA
DESI Collaboration December 8 – 11, 2025
- **DESI July** Berkeley, CA, USA
DESI Collaboration July 8 – 11, 2025
- **DESI December** Cancún, Mexico
DESI Collaboration December 10 – 13, 2024
- **DESI July** Marseille, France
DESI Collaboration July 9 – 12, 2024

- **Fundamental Physics from Future Spectroscopic Surveys** Berkeley, CA, USA
Lawrence Berkeley National Laboratory May 6 – 8, 2024
- **DESI December** Waikoloa, HI, USA
DESI Collaboration December 11 – 14, 2023
- **DESI July** Durham, UK
DESI Collaboration July 17-21, 2023
- **2nd Michigan Cosmology Summer School** Ann Arbor, MI, USA
University of Michigan June 5 – 9, 2023
- **DESI December** Cancún, Mexico
DESI Collaboration December 5 – 9, 2022
- **DESI June** Berkeley, CA, USA
DESI Collaboration June 21 – 24, 2022
- **15th School of Modern Astrophysics** Dolgoprudny, Russia
Moscow Insitute of Physics and Technology July 1 – 12, 2019
- **13th School of Modern Astrophysics** Dolgoprudny, Russia
Moscow Insitute of Physics and Technology July 3 – 21, 2017
- **International School of Subnuclear Physics - 2017** Erice, Italy
"Ettore Majorana" Foundation and Centre for Scientific Culture June 14 – 23, 2017
- **International school on particles, fields and strings** Moscow, Russia
National Research University "High School of Economics" April 17 – 24, 2017
- **Astronomical practice** Nizhniy Arkhyz, Russia
Special Astrophysical Observatory June 25 – July 2, 2016

Outreach and teaching

- **DESI redshift-space distortions animations** with Claire Lamman+
Early version used in [the press-release](#) and following news articles Nov 19, 2024 – Mar 2, 2025
- **Astronomy 200: Radiative Processes in Astrophysics** Harvard University
Teaching Fellow Fall 2023
- **Astronomy 201: Astrophysical Fluids & Plasmas** Harvard University
Teaching Fellow Spring 2023
- **Astronomy 130: Introduction to Cosmology** Harvard College
Teaching Fellow Fall 2022

Awards, grants and honors

Dean's Certificate in Recognition of Outstanding Academic Achievements (TAU)	2019 – 2020
Stipend for excellent students of MIPT in the name of A. Abramov	2016 – 2017
International Physics Olympiad, bronze medal	Mumbai, 2015
International Physics Olympiad, silver medal	Astana, 2014

Languages

- **Russian:** native
- **Ukrainian:** fluent
- **English:** advanced
- **Hebrew:** advanced
- **German:** intermediate